



A spatial panel data approach to estimating U.S. state-level energy emissions



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ABSTRACT

We take advantage of a long panel data set to estimate the relationship between U.S. state-level carbon dioxide (CO₂) emissions, economic activity, and other factors. We specify a reduced-form energy demand model to account for energy consumption activities that drive energy-related emissions. We contribute to the literature by exploring several spatial panel data models to account for spatial dependence between states. Estimation results and rigorous diagnostic analysis suggest that: (1) economic distance plays a role in intra- and inter-state CO₂ emissions; and (2) there are statistically significant, positive economic spillovers and negative price spillovers to state-level emissions.

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1. Introduction

Economists, ecologists, private industries and government decision-makers have long been interested in the relationship between energy consumption, economic growth, and environmental quality. These relationships are often the subject of intense public policy debates such as the discussions surrounding the recent U.N. climate change conference in Durban, South Africa. In the U.S. many opponents to climate change related legislation claim that carbon pollution abatement policies may hinder economic growth. Supporters, on the other hand, claim that such policies are absolutely necessary to prevent irreversible global warming caused by anthropogenic emissions of greenhouse gases.

In order to determine whether abatement policies would be harmful to economic growth, policy makers must first determine whether carbon emissions are indeed related to economic activity. It may seem apparent that emissions and economic activity are inextricably linked yet modern day economists, still as of yet, have not been able to consistently determine a causal relationship between the two. Moreover, there is

still no consensus on the drivers of carbon emissions. We do know that anthropogenic carbon emissions are largely caused by the combustion of fossil fuels. Therefore, an economic model of energy demand would seem to be a good approach to better understand the relationship between emissions and growth. Past studies often examined this relationship across a panel of different countries. Although important for policy implications, empirical analysis at such an aggregated level is unable to capture the complexity of different economies, histories, and environmental policies that are unique to each individual country. In this study we further disaggregate country-level data by exploring a panel of U.S. state-level data of emissions, income, and other covariates.

Two past studies have recognized the importance of analyzing the state-level relationship between emissions and income: Aldy (2005) and Auffhammer and Steinhauser (2007). Aldy (2005) tests the environmental Kuznets curve hypothesis between state-level CO₂ emissions and income. Auffhammer and Steinhauser (2007) further the study of Aldy (2005) by using a spatial econometrics model to forecast state-level COV emissions. Aldy (2005) offers a model to explain emissions but fails to explicitly control for spatial interactions between states; Auffhammer and Steinhauser (2007) control for spatial interactions but do not explore differing data generating processes for spatial dependence, nor do they offer a rigorous interpretation of the spatial impacts. These small deficiencies present a gap in the literature. This paper, therefore, contributes to the literature by extending these previous

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two studies. The main focus of this paper is to determine how spatial dependence affects the drivers of state-level carbon dioxide emissions. Four unique contributions of this paper are: (1) more explicitly considering and testing for the types of spatial dependence within the relationship; (2) using recently developed, spatial panel data models and diagnostics to determine the most appropriate spatial econometric model; (3) offering a more rigorous interpretation of both the direct and indirect (spillovers) spatial impacts; and (4) extending the data set to include the years 2001–2009, which is important for capturing recent developments in state-level energy consumption and economic growth.

Looking ahead, our estimation results suggest five conclusions about the relationship between state-level CO₂ emissions and economic growth. One, economic distance between states has a positive and statistically significant affect on own and neighboring CO₂ emissions. “Economic distance” is a concept that suggests the closer the two regions (states) are in geographic distance to one another, the more likely the economic activity within each region will be affected by one another (Conley and Ligon, 2002). This concept specifically recognizes that economic growth across regions (states) is not independent of the economies of others. Two, economic activity (consistent with the definition of economic distance) in one state has a positive, short-run direct impact on its own CO₂ emissions and a positive, short-run indirect impact on neighboring emissions. Three, increasing electricity and oil prices in one state has a negative, short-run direct impact on its own emissions and a negative, short-run indirect impact on neighboring emissions. Four, additional heating degree days have a positive, short-run direct impact on own emissions and a positive, short run indirect impact on neighboring emissions. Five, estimated elasticities are larger (in absolute terms) for the models with spatial dependence (over the models without spatial interactions) because the elasticities capture interaction effects between neighboring states.

The rest of this paper is structured as follows. Section 2 will offer a conceptual framework to motivate the basic model setup and consider how spatial interactions may affect state-level emissions. In Section 3 we will extend the model to include the spatial interactions in the data generating process and discuss diagnostic tests. Section 4 will provide a description of the data. In Section 5 we will present the empirical model and estimation results. Finally, in Section 6 we will discuss implications, limitations, and suggestions for future research.

2. Conceptual framework

In this paper we analyze the relationship between energy consumption, economic activity, and pollution emissions while controlling for potential spatial effects within the data. The pollution variable, carbon dioxide (CO₂), examined in this paper is estimated by the Department of Energy (DOE) based upon the conversion of fossil fuels to their final energy use; e.g., the conversion of coal into electrical energy in a power plant generates emission gases as a byproduct of the combustion process. In other words, CO₂ emissions are estimated based upon a state's observed energy use. Therefore, CO₂ emissions are not to be confused with actual CO₂ pollution that is emitted from the end of a smokestack or tailpipe.³ As the CO₂ data in this paper do not constitute actual CO₂ emissions we will in general refer to this variable as *energy emissions* to avoid any confusion — a more thorough explanation of the emissions data, including how it is estimated, is provided in the Data description section.

As emissions are estimated from state-level energy use, our analysis is implicitly based upon the relationship between energy consumption (i.e., consumption leads to emissions) and economic activity. Hence, we use a reduced-form energy demand model to explain the difference in state-level energy consumption. According to Ryan and Plouffe (2009), a reduced-form energy demand model is specified as follows

$$\ln E = \beta_1 + \beta_2 \cdot \ln P + \beta_3 \cdot \ln Y + e, \quad (2.1)$$

where E denotes energy consumption; P denotes energy price(s); Y denotes a measure of income or aggregate economic activity; and e denotes a stochastic error term.⁴ All variables are expressed in natural logs to account for growth rates and allow for the estimated coefficients to be interpreted as elasticities. Since emissions are based upon energy consumption we can substitute energy emissions for energy consumption in Eq. (2.1). We will further extend this model by explicitly considering spatial dependence, state-level heterogeneous effects, and time-period effects.

We examine emissions on a per capita basis to control for population growth within each state. A justification for examining per-capita CO₂ emissions is outlined in a recent report by the US Energy Information Administration (2012). According to this report: “It is difficult to compare total carbon dioxide emissions across States because of variation in their sizes. One way to normalize emissions across States is to divide them by State population and examine them on a per capita basis.”

Aldy (2005) used a similar data set as this paper and found mixed evidence for the environmental Kuznets curve hypothesis with CO₂ emissions at the state-level in the contiguous U.S. Aldy made an important distinction between consumption-based emissions and production-based emissions. He argued that through interstate trade, a state's emission intensity from production may differ from its intensity from consumption. To account for this distinction he modified the data for states that are net exporters of electricity by deducting the state's average electricity carbon intensity (as a proxy for exported electricity) from its total emissions for a given year. Carson (2010) points out that this distinction is important because it helps control for net electricity importing states that consume energy without experiencing externalities associated with their production.

Consistent with the insight of linkages between state emissions through commerce (Aldy, 2005; Carson, 2010), we explicitly incorporate a term for spatial dependence to account for linkages between states that may affect intra-state and inter-state CO₂ emissions. In other words, we argue that there is potential spatial dependence between state-level economic activities and state-level energy consumption which in turn creates carbon dioxide emissions.

The idea of spatial dependence in this relationship has been captured by two recent studies: Auffhammer and Steinhauser (2007) and Auffhammer and Carson (2008). These studies use spatial econometric models to forecast CO₂ emissions based upon the relationship between energy consumption, economic growth, and pollution emissions — which the authors claim to exhibit spatial dependence. For example, Auffhammer and Steinhauser (2007) use the same dataset of CO₂ emissions as this study to estimate a variety of short-run forecasts of emissions in the U.S. to compare forecasts of state-level data versus nationally aggregated data. By using the state-level data and controlling for spatial effects, the authors find significant improvement in forecasting performance.

Auffhammer and Carson (2008) use a similar model specification and spatial econometric procedure as Auffhammer and Steinhauser (2007) to forecast province-level CO₂ emissions in China. They find that model selection criteria favor a class of dynamic models with spatial dependence over the static, non-spatial model.

There are in principal three types of spatial dependence that may manifest itself in the relationship between emissions, energy consumption, and economic activity. The first type is a spatial lag model, in which the dependent variable, energy emissions, in state i is affected by the emissions in state j .⁵ Loosely speaking, this specification captures spatial spillovers; in other words, the emissions in one place predict an increased likelihood of similar events in neighboring places. From an air pollution perspective this is arguably the most intuitive spatial process

³ The CO₂ data should not be confused either with atmospheric CO₂ pollution, which following emission enters the upper atmosphere and is more global in scope.

⁴ To make our empirical estimates comparable with the results of Aldy (2005) we can extend Eq. (2.1) to express a quadratic polynomial of income.

⁵ This is the model specification assumed by Auffhammer and Steinhauser (2007) and Auffhammer and Carson (2008), but there are other types of spatial autocorrelation (or dependence) models.

to imagine. However, one must recall that we are not dealing with actual emissions but rather estimates of emissions from energy use. So if one were to make an appeal to a spatial lag process then spatial dependence is not represented by geographic space (i.e., we are not tracking atmospheric emissions traveling across state boundaries) but rather by “economic” distance in which the economic activity that drives energy consumption in one state affects its neighboring state(s).

The second type of spatial dependence involves a spatial lag of the dependent variable and a spatial lag of the explanatory variables; this is referred to as a spatial Durbin model. In this case it is assumed that there is not only spatial dependence within the dependent variable but the determinants of energy demand such as energy prices and income in one state are directly affected by neighboring states.

The third type of spatial dependence is spatial error in which the error terms across different spatial units are correlated. With spatial error in an ordinary least squares regression, the assumption of uncorrelated (independently distributed) errors is violated, and as a result the estimates are inefficient. Spatial error is indicative of omitted variables that if left unattended may affect inference.

We will compare model specifications of each of these three spatial dependence processes against the traditional, non-spatial panel data model.

3. Methodological approach

To control for state-level independent effects we propose the following fixed-effects model:

$$y = X\beta + (\iota_T \otimes I_N)\mu + (I_T \otimes \iota_N)\eta + u, \quad (3.1)$$

where y denotes a $(NT \times 1)$ vector of U.S. state-level per capita carbon dioxide emissions. X is an $(NT \times K)$ matrix of the explanatory variables including energy prices, per capita GDP, per capita GDP squared, and climate variables. All the terms are represented in natural logs to capture growth rates. The coefficient μ denotes the individual effect (or heterogeneity) for each U.S. state and η denotes the time effect. ι_N denotes a column vector of ones of length N , or T if the subscript “T” is indicated. I_N denotes an identity matrix of dimensions $(N \times N)$, or dimensions $(T \times T)$ if the subscript “T” is indicated. In the present analysis we treat the individual effect as fixed meaning that we assume that this variable is correlated with the explanatory variables and approximately “fixed” over time for each state within the sample. If we allow for the fixed effects term to enter into the error term and we estimate Eq. (3.1) without controlling for it, then the estimates will result in omitted variable bias (i.e., if the fixed effect is correlated with the explanatory variables). To control for fixed effects we can either (1) estimate μ and η directly in the model by creating dummy variables for these parameters as in a least squares dummy variable (LSDV) model; or (2) we could demean the data as in a fixed effects (FE) or within estimator. The FE model is usually preferred to the LSDV model due to the incidental parameter problem of having to estimate parameters for the covariates, the fixed effects, and time-period effects terms (Table 1).

An alternative assumption in Eq. (3.1) is that state-level individual effects, μ , are not fixed but rather are unobserved “random” variables which follow a probability distribution (usually normal) with finite parameters — this is referred to as the random effects (RE) estimator. The key difference between these estimators is in the assumption of orthogonality of μ . If μ is assumed to be uncorrelated with the explanatory variables then RE is the appropriate estimator; conversely, if μ is assumed to be correlated with the explanatory variables then FE is the appropriate estimator. The Hausman specification test statistic can be used to determine which estimator provides a better fit of the data (Hausman, 1978; Lee and Yu, 2010b). This test statistic is asymptotically distributed as χ^2 with K degrees of freedom, where K is the number of explanatory variables. Another way to test the random effects estimator against the fixed effects estimator is to estimate the “phi” parameter as outlined in

Table 1
Nomenclature.

δ	Spatial autocorrelation coefficient on the spatially lagged dependent variable
ρ	Spatial autocorrelation coefficient on the spatial error term
γ	Spatial autocorrelation coefficient on the explanatory variables
W_N	$(N \times N)$ matrix of spatial weighting coefficients
μ	State-level fixed effects
η	Time-period effects
$\otimes$	Kronecker product
ι_M	$(M \times 1)$ vector of ones, for $M = N$ or T
I_M	$(M \times M)$ identity matrix, for $M = N$ or T
ϕ	Parameter to test for fixed effects vs. random effects (Baltagi, 2005)
SAR FE	Spatial autoregressive model with fixed effects
SAR RE	Spatial autoregressive model with random effects
SDM	Spatial Durbin model
SEM	Spatial error model

Note: We specify a first-order, binary spatial weighting matrix in which if two states are neighbors then the weight coefficient is equal to one or zero otherwise. The diagonal elements of the weighting matrix equal zero as a state cannot be neighbor to itself.

Baltagi (2005). If this parameter equals zero, the random effects estimator converges to its fixed effects counterpart (Elhorst, 2010).

In this study we will assume that the state-level individual effects are correlated with the explanatory variables, and therefore our primary focus will be on the FE estimation procedure. However, for the sake of completeness we will estimate both the FE and RE models and then use the diagnostic tests discussed in the previous paragraph to determine which model provides a better fit to the data.

Given the assumption of fixed effects, there are many different avenues to estimate the parameters in Eq. (3.1). One, we can assume homogeneity of parameters across states, pool the data, and estimate a single demand equation by ordinary least squares (i.e., omitting the fixed effects and time effects terms). OLS is consistent if the disturbances are orthogonal to the right-hand side (RHS) variables. Two, allow for a limited degree of heterogeneity in time invariant unobservables by adopting a fixed effects estimator — this approach still assumes that all coefficients are identical across states. Three, allow all the coefficients to vary across states. Under this assumption, one could potentially estimate the equations state by state which could result in imprecise estimated coefficients due to the short time series for any given state.

Baltagi and Griffin (1997) explore a large number of estimators, including an instrumental variables estimator, and compare the plausibility of the estimators for a dynamic demand model for gasoline in 18 OECD countries. They found that the pooled estimators yield the most plausible estimates. Therefore, we proceed by assuming homogeneity of the parameters across states, pool the data, and estimate a single demand equation.

We specify a fixed effects model to control for possible endogenous characteristics of the individual states within the study — these are characteristics that do not change (or change very little) over time such as unobservable geographic characteristics. The time-period effects control for time-specific shocks that may affect per capita energy consumption in all states such as oil shocks, recessions, and federal policies applicable to all states (Aroonruengsawat et al., 2012). An example of such a federal policy is the second amendment to the Clean Air Act of 1990.

The contribution of this paper is to not only consider state-level unobserved heterogeneity and time-period effects, but also extend the model in Eq. (3.1) to consider how economic distance may affect this relationship. We introduce spatial effects into the model by using a standard (pre-specified and non-negative) spatial weighting matrix, W_N , as an $(N \times N)$ positive matrix where the rows and columns correspond to the cross-sectional observations (contiguous 48 states). An element of the weighting matrix, w_{ij} , expresses the prior strength of interaction between state i and state j . Since we are dealing with a spatial panel, the weights are extended to the entire panel as

$$W_{NT} = I_T \otimes W_N, \quad (3.2)$$

where I_T again denotes the identity matrix of dimension T and $\otimes$ denotes the Kronecker product.

Following our discussion in the previous section, there are three potential types of spatial panel specifications: the spatial lag, spatial error, and spatial Durbin models. The spatial lag or spatial autoregressive model (SAR) can be expressed in matrix form as

$$y = \delta(I_T \otimes W_N)y + X\beta + (I_T \otimes I_N)\mu + (I_T \otimes I_N)\eta + u, \quad (3.3)$$

where δ is the spatial autoregressive coefficient.⁶ We have suppressed all subscripts for cross-sections (i) and time (t) for ease of exhibition.

The spatial error model (SEM), on the other hand, can be expressed as

$$y = X\beta + (I_T \otimes I_N)\mu + (I_T \otimes I_N)\eta + u \quad (3.4)$$

$$u = \rho(I_T \otimes W_N)u + \epsilon, \quad (3.5)$$

where u reflects the spatially autocorrelated error term and ρ denotes the spatial autocorrelation coefficient on the error term.

The (unrestricted) spatial Durbin model (SDM) is specified as

$$y = \delta(I_T \otimes W_N)y + X\beta + \gamma(I_T \otimes W_N)X + (I_T \otimes I_N)\mu + (I_T \otimes I_N)\eta + u, \quad (3.6)$$

where the parameters are the same as before but the parameter γ now indicates a spatial autocorrelation coefficient on the explanatory variables. The spatial Durbin model can be used to determine if the model can be simplified to a spatial lag model or a spatial error model because the models nest dependence in both the disturbances and the dependent variable (LeSage and Pace, 2009). The two null hypothesis tests for determining the correct spatial model are: $H_0: \gamma = 0$ and $H_0: \gamma + \delta \cdot \beta = 0$ (Elhorst, 2009; LeSage and Pace, 2009). The first hypothesis determines if the spatial Durbin can be simplified to the spatial lag model whereas the second hypothesis determines if it can be simplified to the spatial error model. The second hypothesis stems from the fact that the spatial Durbin model is the reduced form of the spatial error model. If both hypotheses are rejected then the spatial Durbin model provides the best fit for the data. Table 1 offers an outline of the nomenclature used throughout the rest of this manuscript. It is provided to help the reader avoid confusion since we discuss so many different types of parameters estimates, acronyms, and matrix operators.

To further test if a spatial effects model outperforms a model without any spatial interaction effects, one may use Lagrange Multiplier (LM) tests for a spatially lagged dependent variable and for spatial error autocorrelation — these tests contain robust counterparts as well (Debarys and Ertur, 2010).⁷ If a spatial lag model and a spatial error model are estimated separately then likelihood ratio (LR) tests can be conducted to determine which model provides the best fit for the data. The LR tests can also be complemented with Wald tests (Elhorst, 2010).

Finally, after running the regressions and conducting the diagnostic tests (LM, LR, and Wald) to determine the most appropriate model, we further the analysis by one additional step. LeSage and Pace (2009) point out that several past studies use the estimated coefficient on the spatial autocorrelation term to test for the existence of spatial spillovers. However, these authors point out that using the point estimates when comparing spatial models may lead to erroneous conclusions. The authors suggest a procedure for calculating direct and indirect impacts. An interpretation of these impacts is that a change in a single observation (state) associated with any given explanatory variable will affect the state itself (*direct impact*) and potentially affect all the other states

indirectly (*an indirect impact*) (LeSage and Pace, 2009).⁸ The sum of these impacts is defined as the *total impacts*. These coefficients differ from the point estimates because there are feedback effects that arise as a result of impacts passing through neighboring states and then back through the states themselves (LeSage and Pace, 2009). Therefore, we will report both the point estimates and the cumulative effects separately to offer a richer interpretation of spatial spillovers within the model.

4. Data description

The energy emissions data were obtained from the Carbon Dioxide Information Analysis Center (CDIAC) within the U.S. Department of Energy (Blasing et al., 2004). CDIAC estimates the emissions by multiplying state-level coal, petroleum, and natural gas consumption by their respective thermal conversion factors. Therefore, this dataset represents estimates of CO₂ emissions and not actual emissions (hence the reason for calling this variable energy emissions to avoid confusion), which is somewhat problematic as actual emissions data would be more desirable. The reason for using this particular dataset, however, is that it offers emissions estimates dating back to 1960, well before the establishment of the Environmental Protection Agency (EPA) and stronger enforcement of the U.S. Clean Air Act. Energy emissions are offered in per capita terms to control for changes in energy consumption based upon changes in the state population. The energy emission estimates are extended by using more recent calculations of energy-related carbon dioxide emission (2000–2009) offered by the (US EIA, 2012). EIA calculates emissions identically to the CDIAC; however, we visually inspected the data to ensure that new emission estimates are consistent with the previous estimates. The estimates are offered in units of metric tonnes per person.

Itkonen (2012) offers the following simple explanation of how the energy emissions are estimated. The CDIAC and EIA define carbon dioxide emissions as a linear function of fossil fuel combustion and cement manufacturing. The amount of CO₂ emissions is determined by the chemical composition of the fuel source. Emission estimates are calculated by multiplying the amount of fuel usage by a constant thermal conversion factor as determined by the chemical properties of the fuel. Therefore, CO₂ emissions are a linear combination of the per-capita usage of oil, E_t^{oil} , solid fuels such as coal, E_t^{coal} , natural gas, E_t^{gas} , and emissions from cement manufacturing, S_t . Formally, this is expressed as

$$CO_{2,t} \equiv \alpha_{oil} \cdot E_t^{oil} + \alpha_{coal} \cdot E_t^{coal} + \alpha_{gas} \cdot E_t^{gas} + \alpha_{flare} \cdot E_t^{flare} + S_t, \quad (4.1)$$

where α_{oil} , α_{coal} , α_{gas} , $\alpha_{flare} > 0$ are the related thermal conversion factors.

The GDP data was obtained from the Bureau of Economic Analysis (BEA) within the U.S. Department of Commerce (US Bureau of Economic Analysis, 2012). The BEA offers annual state-level GDP estimates from 1963 to the near present. The estimates are based on per capita nominal GDP by state. The estimates were converted to real dollars by using the BEA's implicit price deflator for GDP.

To model climatic influences on energy demand we use Cooling Degree Days (CDD) and Heating Degree Days (HDD), which were obtained from the National Climate Data Center within the National Oceanic and Atmospheric Administration (National Climate Data Center, 2010). CDD (or HDD) is a unit of measure to relate the day's temperature to the energy demand of cooling (or heating) at a residence or place of business—it is calculated by subtracting 65° Fahrenheit from the day's average temperature (Swanson, 2010). Residential energy consumption has been found to be highly correlated with CDD and HDD (Diaz and Quayle, 1980). Since the CO₂ emissions are estimated from energy consumption, the CDD and HDD data as quantitative indices should capture much of the year-to-year variation in energy consumption. CDD and HDD are expected to be positively related to CO₂ emissions as cooler (or hotter) days would induce households or businesses to demand

⁶ If we assume fixed effects then we will denote the model as a spatial autoregressive model with fixed effects (SAR FE). If we assume random effects then we denote this model as SAR RE.

⁷ The LM test statistics only require estimation of a non-spatial model associated with the null hypothesis that the spatial autocorrelation coefficient is equal to zero (LeSage and Pace, 2009).

⁸ A brief theoretical explanation of these impacts is offered in the Appendix; however, a more thorough explanation, including how the impacts are estimated, is provided in Kim et al. (2003) and LeSage and Pace (2009).

higher amounts of energy for heating (or cooling) a residence or place of business.

Energy prices were obtained from the EIA ([US Energy Information Administration, 2008](#)). The energy prices represent state-level annual average prices of coal, natural gas, and oil. The prices were converted to real values by again using the BEA's implicit price deflator – this ensures that the index used to convert nominal to real values is consistent with that of state-level GDP.

Annual state population data were obtained from the [U.S. Census Bureau \(2010\)](#). These population estimates represent the total number of people of all ages within a particular state.

The descriptive statistics for the variables are offered in [Table 2](#).

5. Empirical estimation and results

The empirical model is specified as follows

$$\ln(y_{it}) = \beta_0 + \beta_1 \ln(p_{it}^c) + \beta_2 \ln(p_{it}^{ng}) + \beta_3 \ln(p_{it}^o) + \beta_4 \ln(p_{it}^e) + \beta_5 \ln(GDP_{it}) + \beta_6 \ln(GDP_{it})^2 + \beta_7 \ln(CDD_{it}) + \beta_8 \ln(HDD_{it}) + \beta_9 \ln(y_{it-1}) + \mu_i + \eta_t + u, i = 1, \dots, N; t = 1, \dots, T \quad (5.1)$$

where y_{it} is real per-capita energy emissions in state i at time t . GDP_{it} denotes real per-capita state-level GDP. CDD_{it} denotes cooling degree days, whereas HDD_{it} denotes heating degree days. The variables p_{it}^c , p_{it}^{ng} , p_{it}^o and p_{it}^e denote state-level prices of coal, natural gas, crude oil, and electricity respectively. The scalar parameters on the price yield measures of price elasticities of demand. Consistent with economic theory (the law of inverse demand) we predict all these parameters to be negative. We assume fixed state-specific effects, μ_i , and time-period effects are denoted by η_t . The observations in Eq. (5.1) are available in the 48 contiguous states from 1970–2009 so that $T = 40$ and $N = 48$.⁹

Eq. (5.1) is a reduced-form model for energy demand ([Ryan and Ploure, 2009](#)) with the simple extension of adding the quadratic polynomial expression of GDP, and adding the climatological variables (CDD and HDD). Without the squared term of GDP, the model is very similar in nature to that of [Aroonruengsawat et al. \(2012\)](#). There is a potential problem of price endogeneity in Eq. (5.1) due to an increasing block price structure in electricity – this may lead to an upward-bias in the estimate of demand response ([Hanemann, 1984](#)). A common way to deal with this problem is to estimate marginal prices instead ([Berndt, 1996](#)), but such data is unfortunately unavailable for our sample, so following [Baltagi et al. \(2002\)](#) and [Maddala et al. \(1997\)](#) we will assume price exogeneity. We also abstract away from issues related inter-fuel substitution which have implications for price interaction effects.

In order to determine which type of model (spatial vs. non-spatial) best fits the data, we begin our investigation by testing several different model specifications. This testing procedure is a mixture between a specific-to-general approach and general-to-specific approach ([Elhorst, 2010](#)). The procedure begins by testing the non-spatial model against the spatial lag and spatial error models. If the non-spatial models are rejected, the spatial Durbin model is tested to determine if it can be simplified to either the spatial lag or spatial error model – this step seeks corroborating evidence from the first step.

The estimation results for the non-spatial panel data models are reported in [Table 3](#). Columns (1)–(4) represent estimation results given the specification of: pooled OLS (no fixed or time-period effects), fixed effects only (no time-period effects), time-period effects only (no fixed effects), and both fixed effects and time-period effects, respectively.

To investigate the null hypothesis that the fixed effects and time-period effects are jointly insignificant, we performed a likelihood ratio test. The LR test for the joint insignificance of the fixed effects was rejected at the one percent level (4347.0321, 48 degrees of freedom,

Table 2
Descriptive statistics.

Variables	Max	Min	Mean	Median	Std dev
CO ₂	131.11	7.71	23.72	19.46	16.49
Coal price	7.86	0	2.19	1.99	0.9425
Elec price	49.43	7.63	25.22	24.11	7.36
Nat gas price	15.87	1.18	6.80	6.54	2.63
Oil price	24.62	5.43	11.73	10.56	3.54
GDP	64,576	13,483	30,050	28,522	8793
CDD	3875	80	1085	867	779.38
HDD	10,745	400	5243	5381	2049.2

Note: CO₂ and GDP represent per-capita values. CO₂ is measured in metric tonnes. GDP and the price data are measured in real USD. Prices are measured as USD per unit of btu.

$p < 0.01$). Likewise, the LR test for the joint insignificance of the time-period effects was also rejected (182.4867, 40 degrees of freedom, $p < 0.01$). These results justify the extension of the model with fixed effects and time-period effects.

Recall that if the state-level fixed effects term is correlated with the explanatory variables but it is not controlled for within the model then OLS estimates will result in omitted variable bias (OVB). The pooled OLS estimates (column (1) in [Table 3](#)) for all the coefficients in the model are all highly statistically significant ($p < 0.001$) which arguably results from the OVB. Given the joint significance of the fixed and time-period effects from the LR test we focus on the estimation results in Column (4) in [Table 3](#). The estimated coefficient on electricity and oil prices conforms with expectations as the coefficients are negative and indicative of inelastic response of emissions to price changes. The coefficient on gas prices does not conform with expectations but this coefficient is only marginally significant. The coefficient on heating degree days is positive (consistent with expectations) and highly statistically

Table 3
Estimation results without spatial interaction effects.

Dependent variable: CO ₂	(1)	(2)	(3)	(4)
Intercept	−59.2637*** (−5.9565)			
Coal price	−0.2840*** (−9.3775)	0.0095 (0.7603)	−0.4247*** (−14.0733)	0.0202 (1.4637)
Electricity price	−0.3198*** (−8.3752)	−0.3506*** (−16.7574)	−0.2339*** (−6.4689)	−0.2600*** (−11.2087)
Gas price	−0.2695*** (−7.3393)	0.0529*** (3.8774)	−0.4332*** (−10.3009)	0.0347* (1.6526)
Oil price	0.2288*** (5.3271)	−0.0269* (−1.7390)	−1.9628*** (−16.6450)	−0.3376*** (−5.5978)
CDD	0.2389*** (11.6404)	0.0283* (1.7703)	0.2291*** (12.5020)	0.0008 (0.0435)
HDD	0.2906*** (9.9685)	0.1281*** (3.4137)	0.3889*** (14.3919)	0.1654*** (3.6262)
GDP	11.5203*** (5.9809)	8.1988*** (12.4428)	0.8412 (0.4202)	7.5689*** (10.4054)
GDP ²	−0.5579*** (−5.9772)	−0.4009*** (−12.5228)	−0.0444 (−0.4585)	−0.3799*** (−10.8477)
σ ²	0.1493	0.0132	0.1157	0.0120
R ²	0.3319	0.2099	0.4780	0.1997
FE R ²		0.9408	0.4820	0.9462
Log-likelihood	893.8655	1432.8	649.5129	1524.0
LM spatial lag	5.6619**	241.9989***	0.08413	138.6038***
LM spatial error	21.8779***	206.9125***	3.6114*	125.0031***
Robust LM	28.0865***	35.5800***	15.3873***	13.7217***
spatial lag				
Robust LM	44.3025***	0.4936	18.9146***	0.1210
spatial error				

Note: All variables are in natural logarithms. The estimates above are based on the following models: (1) pooled OLS (no fixed or time-period effects), (2) fixed effects only, (3) time-period effects only, and (4) fixed and time period effects. The test statistic for the LM test is based on a chi-squared distribution with one degree of freedom.

* Denotes $p < 0.1$.

** Denotes $p < 0.5$.

*** Denotes $p < 0.01$.

⁹ The data set was limited to the years 1970–2009 because those are the only years for which the EIA has data on average annual, state-level energy prices.

significant. The coefficients on the polynomial of income terms are also highly statistically significant and conform with the EKC hypothesis of the inverted U-shaped relationship between emissions and income.

All of the non-spatial models may suffer from misspecification if spatial dependence exists within the data. To test for the presence of spatial dependence we begin by conducting the classical LM tests (Anselin et al., 2008; Burridge, 1980). These test results are listed in the bottom part in Table 3. For the classical LM tests (labeled “LM spatial lag” and “LM spatial error”), the hypothesis of no spatially lagged dependent variable and the hypothesis of no spatially autocorrelated error term are strongly rejected for each of the specifications with the exception of model including time-period effects alone (although the hypothesis of no spatially autocorrelated error is rejected at the ten percent significance level with this specification). Examining these tests’ robust counterparts (Debarsy and Ertur, 2010), both hypotheses are rejected for the pooled OLS and time-period effects specification, but only the hypothesis of no spatially lagged dependent variable can be rejected when fixed effects are introduced into the specification (i.e., columns (2) and (4)). These results imply that a model specification with a spatially lagged dependent variable may be favored over a non-spatial model since we find consistent rejection of the hypothesis of no spatially lagged dependence. A model specification with spatially autocorrelated error terms is questionable as the LM tests offer mixed results.

To further test which spatial model specification is appropriate we estimate a spatial Durbin model and then conduct likelihood ratio tests (Burridge, 1981) as outlined in the Methodological approach section. The results for the spatial Durbin model are listed in Column (1) in Table 4. These results reflect estimation of the SDM with the

Table 5

Post diagnostic tests of spatial specification.

Test	Chi-squared statistic	p-Value
LR		
Spatial lag	10.2658	0.2469
Spatial error	17.6215	0.0243
Wald test		
Spatial lag	10.4900	0.2323
Spatial error	16.5544	0.0351
Hausman	190.0861	0.0000

Note: LR tests follow a chi-squared distribution with K degrees of freedom (Burridge, 1981).

bias correction (Elhorst, 2010; Lee and Yu, 2010a) – results without the bias correction are almost identical. The LR and Wald test statistics are reported in Table 5, and as noted by LeSage and Pace (2009) yield similar results. According to these tests, the first hypothesis ($H_0: \gamma = 0$) cannot be rejected, which suggests that the spatial lag model is the most appropriate specification. The test results indicate that the second hypothesis ($H_0: \gamma + \delta \cdot \beta = 0$) is rejected at the five percent level, which implies that the spatial error model is not appropriate. These LR and Wald test results are consistent with our LM test results in Table 3, which indicates that the spatial lag model is the most appropriate specification for this relationship. Therefore, we report the results for the spatial lag model with fixed and time-period effects (SAR FE) in column (2) in Table 4. For completeness we also report the estimation results for the spatial autoregressive model with random and time-period effects (SAR RE) in column (3).

To test the assumption of whether fixed effects provide a better fit to the data than random effects we conduct a Hausman specification test. The results of the Hausman test are listed in Table 5 – this test suggests that the random effects are rejected at a one percent significance level. In addition to the Hausman test, we estimate the “phi” parameter from Baltagi (2005). If this parameter equals 0, the random effects model converges to its fixed effects counterpart (Elhorst, 2010) – this parameter is reported in Table 4. We find that $\phi = 0.0416$ and is significant at a one percent level, which corroborates to the Hausman test and suggests that the fixed effects assumption is the appropriate specification given the data.

The estimated coefficient on the spatial lagged term of energy emissions is positive, similar in magnitude, and highly statistically significant for both the SDM (column (1) in Table 4) and the SAR models (columns (2) and (3) in Table 4). Given our normalization of the spatial weighting matrix, the significant spatial autocorrelation coefficient suggests that energy emissions in neighboring states, on average, are exerting a positive effect local energy emission. Since energy emissions are estimated based on energy consumption across states, this coefficient implies that economic distance across states is affecting local energy emissions or vice versa. Economic distance could potentially represent energy commerce across states such as the exportation of electricity across state lines. This is consistent with Aldy (2005) and Carson (2010) who argued that interstate energy commerce affects the emissions–growth relationship. Aldy (2005) dealt with interstate electricity commerce by modifying the data for states that are net exporters of electricity. Our approach is somewhat different as we allow for spatial dependence to indirectly account for spillovers due to economic distance.

The estimated coefficients for the energy price terms are all between zero and one (in absolute terms) which indicates that energy emissions are price inelastic in the short run. This is consistent with expectations as in the short run consumers do not have the possibility to change their capital stock (which use these energy inputs) and can only change consumption behavior (Bhattacharyya, 2011). Ergo, price elasticities are expected to be more inelastic in the short run than in the long run as consumers can change their capital stock in reaction to higher energy

Table 4

Estimation results with spatial interaction effects.

Dependent variable: CO ₂	(1)	(2)	(3)
δ	0.3586*** (13.4472)	0.3586*** (14.1340)	0.3940*** (15.5956)
Coal price	0.0101 (0.7209)	0.0120 (0.8940)	0.0050 (0.4179)
Electricity price	−0.1964*** (−7.0207)	−0.2040*** (−8.9620)	−0.2264*** (−10.9728)
Gas price	−0.0324 (−1.3058)	−0.0111 (−0.5409)	0.0700*** (5.5626)
Oil price	−0.2200*** (−3.2676)	−0.2418*** (−4.1076)	−0.0518*** (−3.6709)
CDD	0.0365 (1.0361)	0.0056 (0.3075)	0.0015 (0.0993)
HDD	0.1300 (1.4936)	0.1290*** (2.9060)	0.0506 (1.4502)
GDP	7.0360*** (8.9879)	7.0278*** (9.9384)	0.4680*** (6.4000)
GDP ²	−0.3540*** (−9.4569)	−0.3519*** (−10.3323)	−0.0254*** (−6.4619)
ϕ			0.0416*** (6.9339)
W * Coal price	−0.0022 (−0.0818)		
W * Elec price	−0.0076 (−0.1637)		
W * Gas price	0.0975** (2.4776)		
W * Oil price	−0.0120 (−0.1089)		
W * GDP	−1.7387 (−1.2881)		
σ^2	0.0113	0.0113	0.0123
R ²	0.9516	0.9513	0.9445
Log-likelihood	1594.5733	1589.4405	1585.7626

Note: All variables are in natural logarithms. The estimates above are based on the following models: (1) SDM, (2) SAR FE, and (3) SAR RE.

* Denotes $p < 0.1$.

** Denotes $p < 0.5$.

*** Denotes $p < 0.01$.

prices.¹⁰ In accordance with the law of demand it is expected that all coefficients on the prices are negative — this expectation is violated in the case of coal prices across the different models; however, none of these estimates are statistically significant. According to the SAR FE model results, the estimated coefficients on electricity prices are highly significant across all models and imply that an increase in price by 10% would lead to an approximate decrease in energy emissions by 2.0%; this differs from the estimated elasticity of 2.6% reported in column (4) in Table 3. The estimated coefficients on oil prices are also highly significant across all the models. These coefficients imply that an increase in crude oil prices by 10% would lead to an approximate 2.4% decrease in energy emissions; again, this differs from the estimated elasticity of 3.4% found with the non-spatial.

The estimated coefficients on the climate variables, cooling degree days and heating degree days, are fairly consistent across the three models. The estimated coefficient on cooling degree days is not significant for any of the models, but the positive sign is consistent with expectations as additional hotter days within a year are expected to increase the demand for cooling commercial spaces or private places of residence. The estimated coefficient on heating degree days is highly significant for the SAR FE model. The coefficient estimate from this model implies that an additional 10% increase of cooler days within the year leads to an approximate 1.3% increase in energy emissions; this differs from the estimate 1.7% found with the non-spatial model (column (4) in Table 3). This is consistent with intuition as the additional heating of homes or office spaces requires more electricity, natural gas, or heating fuel — which in turn creates additional emissions. As with the energy prices, the estimated coefficients on the climate variables represent short-run values.

Finally, the estimated coefficients on the quadratic polynomial of GDP are all highly significant and consistent with the inverted U-shaped relationship espoused by the environmental Kuznets curve hypothesis. The signs and magnitudes on these terms are highly consistent across the three specifications. The estimated turning point for energy emissions is approximately \$21,709 (USD) according to the results of the SAR FE model — this puts the turning point somewhere below the median value of per-capita GDP in our sample.¹¹ The estimated turning point given the non-spatial model estimates is very similar at approximately \$21,200 (USD). It is difficult to estimate the year the turning point may have occurred because there is significant variation in per-capita GDP between states, but the U.S. national average of per-capita GDP (real dollars) reached \$22 K between 1988 and 1989.

As pointed out in the Methodological approach section, it is tempting to use the point estimates in Table 4 to test the hypothesis as to whether spatial spillovers exist. The spatial autocorrelation coefficient is positive and statistically significant across all three spatial model specifications as expected — several past studies would stop here and use these estimates as a test for the existence of spatial spillovers. However, a better interpretation of spatial spillovers is reflected in the coefficient estimate of indirect impacts (labeled as *indirect*) presented in Table 6. The difference between the point estimates in Table 4 and the results presented in Table 6 are partially due to the estimated coefficient of the spatially lagged dependent variable (δ in Table 4) and partially due to the estimated coefficients on the independent variables. The presentation of models in Table 6 exactly corresponds with Table 4.

Since the diagnostic tests results suggest that the spatial autoregressive model with fixed effects provides the best fit we will limit our interpretation to these estimates (column (2) in Table 6). A comparison of the total impact estimates in column (2) in Table 6 to the corresponding non-spatial point estimates in

column (4) in Table 3 reveals that all of the coefficient estimates (in absolute terms) are fairly consistent but slightly larger for the model with the spatial autocorrelation. These differences arise because the non-spatial model is potentially misspecified due to the presence of spatial dependence within the data. Another way to interpret this is that the differences are reflected by the feedback effects that arise as a result of impacts passing through neighboring states and back to the state itself. From a policy standpoint, the non-spatial model results may lead policy makers to believe that adopting particular measures, such as increasing electricity prices, will result in smaller reductions in energy emissions. For example, the estimated coefficient on electricity prices in the non-spatial model indicates a short-run, price elasticity of -0.26 ; however, the total impact estimate from the spatial model yields a coefficient of -0.32 . The results of the non-spatial model suggest that 10% increase in oil prices would reduce energy emissions by 2.6% whereas the spatial model suggests a larger reduction of 3.2%.

Comparing the point estimates of the SAR FE model (column (2)) in Table 4 to the impact estimates in Table 6, one will notice that the estimated coefficient on the price elasticity of demand for electricity (in absolute terms) is slightly higher for the direct impact estimate of 0.2110 (Table 6) than the point estimate of 0.2040 (Table 4). The higher coefficient estimate for the direct impact implies that own energy emissions are slightly more responsive to an increase in own electricity prices than is found with the point estimates of the SAR FE. In other words, if one state increases its electricity prices then it has a negative, short-run effect on its own energy emissions. Moreover, an increase in own electricity prices has a negative short-run effect on neighboring states as well. The affect on neighboring states is represented by the *indirect* impact coefficient — for electricity prices this is -0.1073 . An interpretation of this coefficient is that if a state increases its own electricity prices by 10% then it will decrease energy emissions in neighboring states by approximately 1.1% — this is a more appropriate definition of spatial spillovers. We find similar results with the price elasticity of demand for oil prices, although the indirect impact is larger for oil prices; i.e., an increase in oil prices by 10% leads to a decrease in neighboring energy emissions by 1.3%. Further, having additional cooler days throughout the year, as reflected in the estimates of HDD, has a positive direct and a small but positive indirect impact. The coefficient on the indirect impact suggests that a 10% increase in heating degree days leads to a short-run increase of neighboring state's emissions by approximately 0.7%. Finally, the estimates on the gross domestic product term suggest both positive direct and positive indirect effects (both are highly statistically significant) on energy emissions. One will notice once again that the direct impact (7.2854 in Table 6) is higher than the point estimate (7.0278 in Table 4). Unlike the other variables though, the coefficient estimates on income presented here cannot be interpreted as elasticities because we specify a quadratic polynomial of income. Although, using the coefficient estimates of the total impacts implies that the turning point in emissions occurs at slightly high per-capita income of \$21,820 than found with the point estimates of the non-spatial model.

6. Implications and conclusions

Issues associated with spatial dependence have largely been ignored in the energy pollution, energy consumption, and economic growth literature. In this paper, we introduced a spatial panel data model approach to account for spatial dependence that is expected to be found within energy emissions and state-level economic activity. Based on the estimation results from the spatial panel data models, our findings suggest that spatial spillovers indeed affect this relationship between U.S. states.

As our specification is based on a reduced-form model, it is difficult for us to make any appeals to policy implications. We would need a complete model of energy demand and supply in order to appeal to specific policies though our estimation results have indirect implications

¹⁰ We do not explore long-run price elasticities in this particular study because such analyses require dynamic panel data models. The development of dynamic, spatial panel data models is still in its infancy so we abstract away from long-run elasticities.

¹¹ The turning point is calculated by $\exp(-\beta_1/2 \cdot \beta_2)$, where β_1 is the coefficient on the log of GDP and β_2 is the coefficient on the log of GDP, whole quantity squared.

Table 6

A comparison of cumulative impacts from spatial Durbin model, spatial autoregressive fixed effects, and spatial autoregressive random effects.

Coefficient & effect	(1)		(2)		(3)	
Coal price						
Direct	0.0105	(0.7565)	0.0122	(0.8985)	0.0056	(0.4622)
Indirect	0.0016	(0.0429)	0.0062	(0.8913)	0.0033	(0.4622)
Total	0.0121	(0.2863)	0.0184	(0.8979)	0.0090	(0.4629)
Elec price						
Direct	−0.2036***	(−7.3741)	−0.2110***	(−8.9142)	−0.2363***	(−11.0615)
Indirect	−0.0967	(−1.5818)	−0.1073***	(−6.9351)	−0.1374***	(−8.0892)
Total	−0.3003***	(−4.9451)	−0.3182***	(−8.8029)	−0.3737***	(−10.8615)
Gas price						
Direct	−0.0262	(−1.1266)	−0.0121	(−0.5768)	0.0731***	(5.4650)
Indirect	0.1222**	(2.2714)	−0.0063	(−0.5758)	0.0425***	(4.8619)
Total	0.0961*	(1.6954)	−0.0184	(−0.5773)	0.1156***	(5.3950)
Oil price						
Direct	−0.2290***	(−3.3883)	−0.2521***	(−4.2390)	−0.0546***	(−3.8413)
Indirect	−0.1152	(−0.7655)	−0.1280***	(−3.9991)	−0.0317***	(−3.6704)
Total	−0.3442***	(−2.1232)	−0.3801***	(−4.2457)	−0.0864***	(−3.8438)
CDD						
Direct	0.0337	(1.0068)	0.0051	(0.2675)	0.0022	(0.1392)
Indirect	−0.0467	(−0.9240)	0.0026	(0.2653)	0.0013	(0.1393)
Total	−0.0131	(−0.3605)	0.0077	(0.2668)	0.0035	(0.1394)
HDD						
Direct	0.1377**	(1.6917)	0.1338***	(2.9315)	0.0509	(1.4106)
Indirect	0.0526	(0.3997)	0.0680***	(2.8213)	0.0296	(1.3800)
Total	0.1903**	(2.0657)	0.2018***	(2.9265)	0.0805	(1.4043)
GDP						
Direct	7.0711***	(9.2450)	7.2854***	(10.0073)	0.4910***	(6.5094)
Indirect	0.7574	(0.4179)	3.7082***	(6.9274)	0.2850***	(6.0200)
Total	7.8285***	(3.8904)	10.9935***	(9.4424)	0.7760***	(6.6239)
GDP ²						
Direct	−0.3556***	(−9.7006)	−0.3646***	(−10.3971)	−0.0266***	(−6.5575)
Indirect	−0.0348	(−0.3955)	−0.1856***	(−7.0687)	−0.0154***	(−6.0054)
Total	−0.3904***	(−3.9965)	−0.5502***	(−9.7805)	−0.0420***	(−6.6493)

Note: Numbers in parentheses denote corresponding *t*-statistics. The estimates above are based on the following models: (1) SDM, (2) SAR FE, and (3) SAR RE.* Denotes $p < 0.1$.** Denotes $p < 0.5$.*** Denotes $p < 0.01$.

for national and state carbon dioxide policies, especially for policies related to CO₂ abatement with energy emissions. For example, our spatial panel results imply that certain impacts, such as increasing electricity prices, will have a larger effect on mitigating emissions than is suggested by the non-spatial models. Policy makers can use these findings to further our heuristic understanding of the complex nature of the relationship among energy consumption, economic activity, and energy emissions. If the true data generating process for the energy emissions–growth relationship is characterized by spatial dependence then this may signal to policy makers that energy emissions may be an interstate problem. This may be a problem if states engage in electricity commerce and intrastate pollution regulations are driving some states to be pollution havens — this is something we do not explore in this paper. In terms of a practical application, policy makers can use this research to further investigate the spatial qualities of ambient pollution emissions (such as sulfur dioxide emissions) and in turn determine if there are indeed interstate problems; i.e., upwind emitting states driving emissions in downwind states.

This study may have future implications for analyses regarding spatial dependence in energy emissions. For example, [Kindle and Shawhan \(2011\)](#) examine the environmental impact of a regional greenhouse gas initiative (RGGI) among ten states in the northeastern U.S. These ten states allow for electricity to be imported from neighboring states — both states participating in the RGGI and states not participating. The RGGI institutes a tradable permits program to allow for CO₂ emissions in the electricity sector — these allowances, in turn, increase the marginal cost per mega watt-hour for electrical generating companies. If the cost of compliance for the RGGI causes these participating states to import electricity from non-participating states then this may cause inter-regional emission leakages ([Kindle and Shawhan, 2011](#)). These leakages could potentially counteract emission reductions gained in the RGGI

states. The point here is that the relationship between emissions and income in one state may be affected by neighboring states.

Our estimation procedures suffer from some limitations including the problem of measurement error. Our dependent variable of energy emissions is based upon the estimation of carbon dioxide emissions from the combustion of fossil fuels and not actual ambient carbon dioxide emissions. The downside of this variable is that it may be subject to measurement error. The upside of this variable, however, is that it gave us a relatively large dataset along the time dimension which is important for making arguments of asymptotics.

An additional problem is that we specified a single equation, reduced-form model. Although these reduced-form models are used fairly frequently in the energy literature, they offer limited information for policy decisions because they ignore issues of inter-fuel substitution, technical change, and changes in supply ([Bhattacharyya, 2011](#)).

Future research may consider variables that indicate the likelihood of a state adopting voluntary cutbacks in energy emissions. [Auffhammer and Steinhauser \(2007\)](#) consider a measure of the median “green” voter by state offered by the League of Conservation Voters Scorecard indicator. Each year the League of Conservation Voters measures environmental legislation in the U.S. Congress by recording legislator voting behavior by state. Due to data limitations we did not explore this variable within our paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.eneco.2013.07.021>.

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